

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")

Libraries imported successfully!
```

◆ Gemini

```
df = pd.read_csv("googleplaystore.csv")
display(df.head())
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Version
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varie wi devic
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1

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df = pd.read_csv("googleplaystore.csv")

df.head()
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Start coding or [generate](#) with AI.

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df.head()
```

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Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(10841, 13)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10841 entries, 0 to 10840
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   App              10841 non-null  object
1   Category         10841 non-null  object
2   Rating           9367 non-null   float64
3   Reviews          10841 non-null  object
4   Size             10841 non-null  object
5   Installs         10841 non-null  object
6   Type             10840 non-null  object
7   Price            10841 non-null  object
8   Content Rating   10840 non-null  object
9   Genres           10841 non-null  object
10  Last Updated     10841 non-null  object
11  Current Ver      10833 non-null  object
12  Android Ver      10838 non-null  object
dtypes: float64(1), object(12)
memory usage: 1.1+ MB
```

```
df.isnull().sum()
```

	0
App	0
Category	0
Rating	1474
Reviews	0
Size	0
Installs	0
Type	1
Price	0
Content Rating	1
Genres	0
Last Updated	0
Current Ver	8
Android Ver	3

dtype: int64

```
df['Rating'] = df['Rating'].fillna(df['Rating'].median())
```

```
df.isnull().sum()
```

	0
App	0
Category	0
Rating	0
Reviews	0
Size	0
Installs	0
Type	1
Price	0
Content Rating	1
Genres	0
Last Updated	0
Current Ver	8
Android Ver	3

dtype: int64

```
df['Type'] = df['Type'].fillna(df['Type'].mode()[0])
```

```
df.isnull().sum()
```

	0
App	0
Category	0
Rating	0
Reviews	0
Size	0
Installs	0
Type	0
Price	0
Content Rating	1
Genres	0
Last Updated	0
Current Ver	8
Android Ver	3

dtype: int64

df.duplicated().sum()

np.int64(483)

df = df.drop_duplicates()

df.shape

(10358, 13)

df.isnull().sum()

	0
App	0
Category	0
Rating	0
Reviews	0
Size	0
Installs	0
Type	0
Price	0
Content Rating	1
Genres	0
Last Updated	0
Current Ver	8
Android Ver	3

dtype: int64

df.dtypes

	0
App	object
Category	object
Rating	float64
Reviews	object
Size	object
Installs	object
Type	object
Price	object
Content Rating	object
Genres	object
Last Updated	object
Current Ver	object
Android Ver	object

dtype: object

df.info()

```
<class 'pandas.core.frame.DataFrame'>
Index: 10358 entries, 0 to 10840
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   App              10358 non-null  object
1   Category         10358 non-null  object
2   Rating           10358 non-null  float64
3   Reviews          10358 non-null  object
4   Size             10358 non-null  object
5   Installs         10358 non-null  object
6   Type             10358 non-null  object
7   Price            10358 non-null  object
8   Content Rating   10357 non-null  object
9   Genres           10358 non-null  object
10  Last Updated     10358 non-null  object
11  Current Ver      10350 non-null  object
12  Android Ver      10355 non-null  object
dtypes: float64(1), object(12)
memory usage: 1.1+ MB
```

df['Reviews'] = pd.to_numeric(df['Reviews'], errors='coerce')

df['Reviews'].dtype

dtype('float64')

```
df['Installs'] = pd.to_numeric(
    df['Installs'].astype(str).str.replace(',', ' ').str.replace('+', ' '),
    errors='coerce'
)
```

df['Installs'].dtype

dtype('float64')

df['Price'].unique()[:30]

```
array(['0', '$4.99', '$3.99', '$6.99', '$1.49', '$2.99', '$7.99', '$5.99',
      '$3.49', '$1.99', '$9.99', '$7.49', '$0.99', '$9.00', '$5.49',
      '$10.00', '$24.99', '$11.99', '$79.99', '$16.99', '$14.99',
      '$1.00', '$29.99', '$12.99', '$2.49', '$10.99', '$1.50', '$19.99',
      '$15.99', '$33.99'], dtype=object)
```

```
df['Price'] = pd.to_numeric(
    df['Price'].astype(str).str.replace('$', '', regex=False).str.replace(',', '', regex=False),
    errors='coerce'
)
```

```
df['Price'].dtype
```

```
dtype('float64')
```

```
df['Price'].isnull().sum()
```

```
np.int64(1)
```

```
df['Price'] = df['Price'].fillna(df['Price'].median())
```

```
df['Price'].isnull().sum()
```

```
np.int64(0)
```

```
df['Installs'] = pd.to_numeric(
    df['Installs'].astype(str)
    .str.replace(',', '', regex=False)
    .str.replace('+', '', regex=False),
    errors='coerce'
)
```

```
df['Installs'].dtype
```

```
dtype('float64')
```

```
df['Installs'].isnull().sum()
```

```
np.int64(1)
```

```
df['Installs'].isnull().sum()
```

```
np.int64(1)
```

```
df['Installs'] = df['Installs'].fillna(df['Installs'].median())
```

```
df[df['Installs'].isnull()]
```

App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver	
-----	----------	--------	---------	------	----------	------	-------	----------------	--------	--------------	-------------	-------------	--

```
df['Installs'].isnull().sum()
```

```
np.int64(0)
```

```
df.shape
```

```
(10358, 13)
```

```
df.describe()
```

	Rating	Reviews	Installs	Price
count	10358.000000	1.035700e+04	1.035800e+04	10358.000000
mean	4.205165	4.059046e+05	1.415640e+07	1.030701
std	0.506868	2.696778e+06	8.023580e+07	16.277843
min	1.000000	0.000000e+00	0.000000e+00	0.000000
25%	4.100000	3.200000e+01	1.000000e+03	0.000000
50%	4.300000	1.680000e+03	1.000000e+05	0.000000
75%	4.500000	4.641600e+04	1.000000e+06	0.000000
max	19.000000	7.815831e+07	1.000000e+09	400.000000

df['Category'].value_counts().head(10)

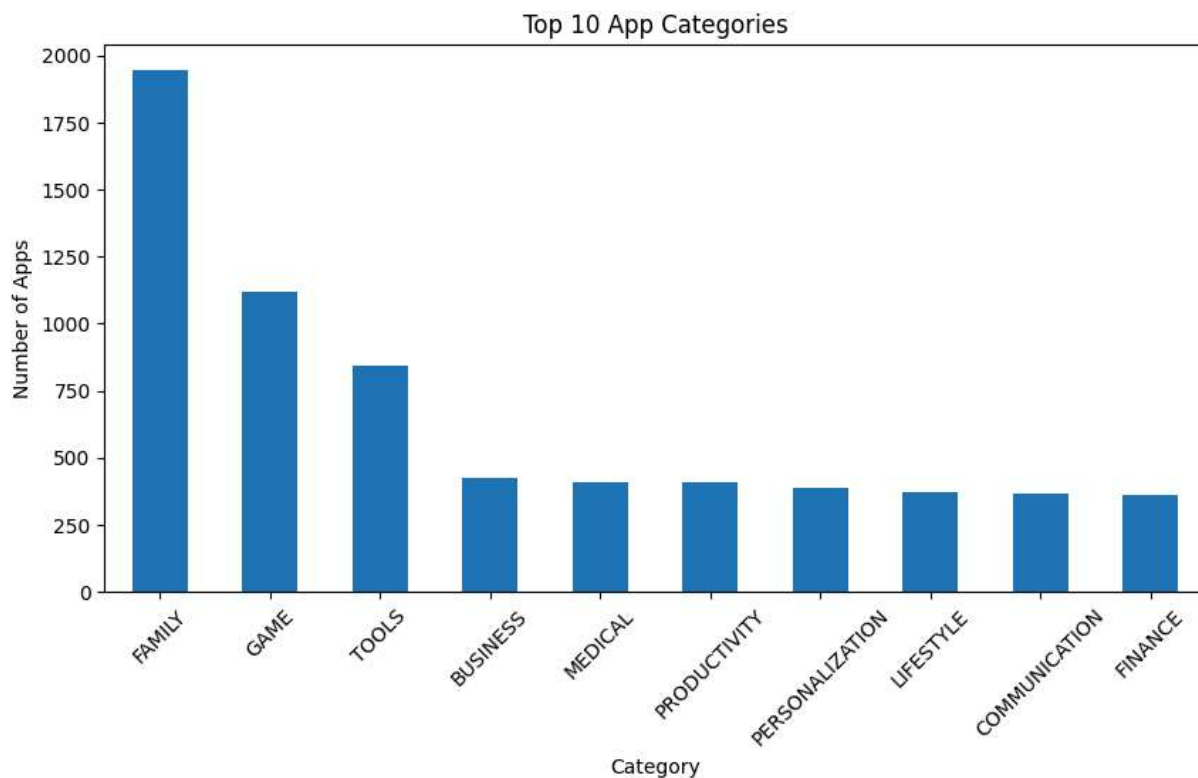
	count
Category	
FAMILY	1943
GAME	1121
TOOLS	843
BUSINESS	427
MEDICAL	408
PRODUCTIVITY	407
PERSONALIZATION	388
LIFESTYLE	373
COMMUNICATION	366
FINANCE	360

dtype: int64

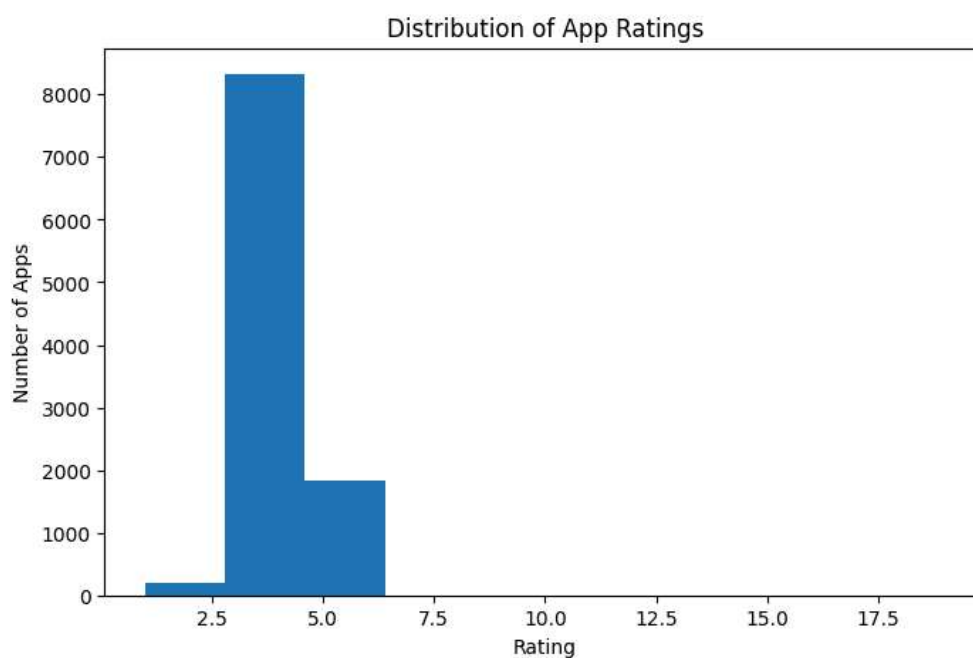
df[['App', 'Rating']].sort_values(by='Rating', ascending=False).head(10)

	App	Rating
10472	Life Made WI-Fi Touchscreen Photo Frame	19.0
8582	Disciple Maker's (DM) Lab	5.0
8572	DM Buddy » Learn Digital Marketing	5.0
5685	Jobs in Canada - Emplois au Canada	5.0
8526	DL Image Manager	5.0
5763	Tozer Devotional -Series 1	5.0
10659	FN	5.0
6146	ReactNative BG Geolocation	5.0
8601	DN Calculators	5.0
8596	DN Employee	5.0

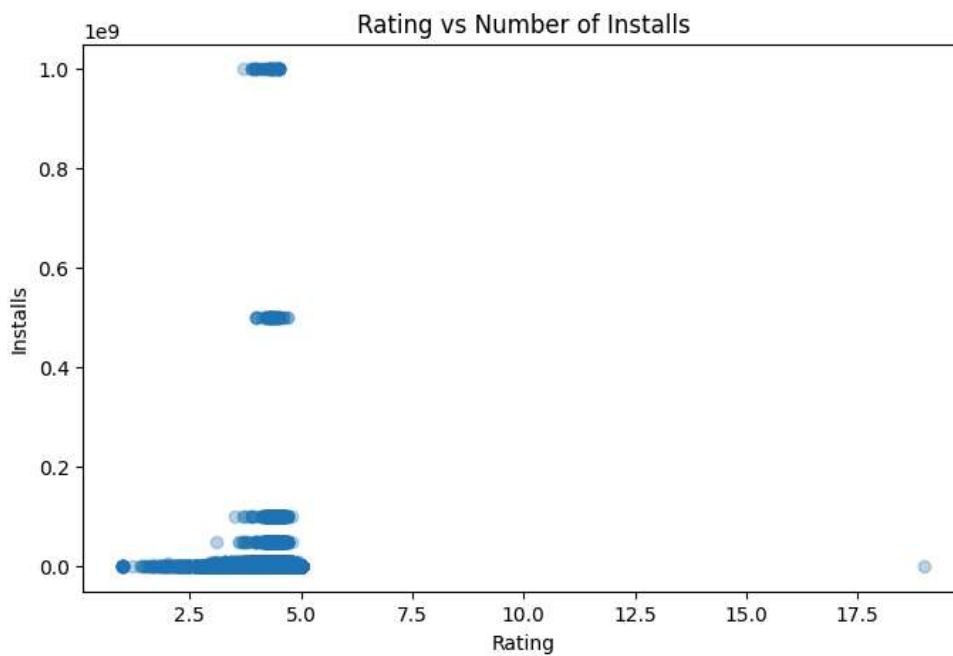
```
df['Category'].value_counts().head(10).plot(kind='bar', figsize=(10,5))
plt.title('Top 10 App Categories')
plt.xlabel('Category')
plt.ylabel('Number of Apps')
plt.xticks(rotation=45)
plt.show()
```



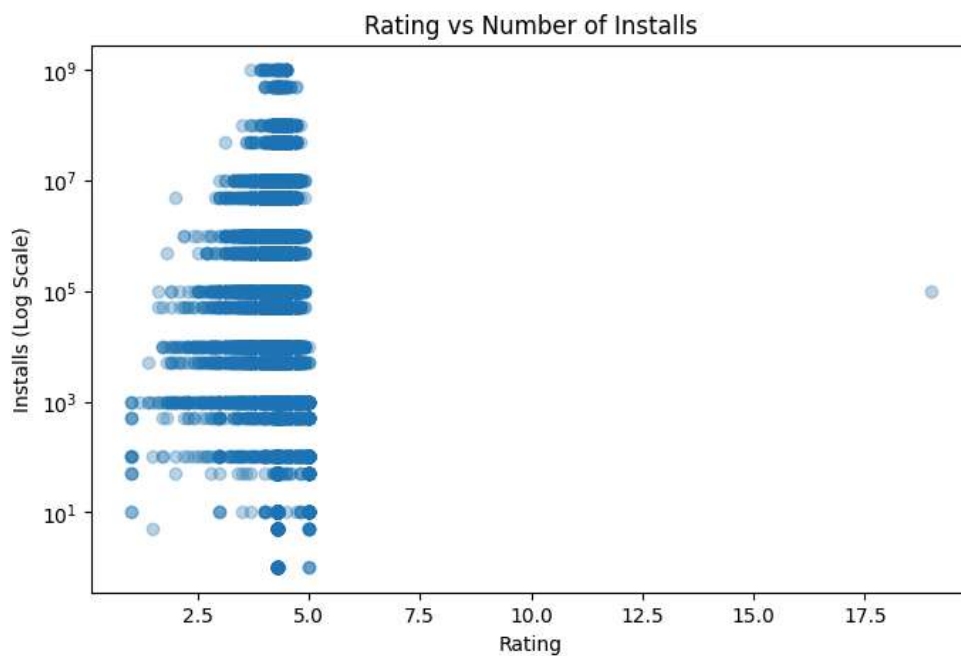
```
plt.figure(figsize=(8,5))
plt.hist(df['Rating'], bins=10)
plt.title('Distribution of App Ratings')
plt.xlabel('Rating')
plt.ylabel('Number of Apps')
plt.show()
```



```
plt.figure(figsize=(8,5))
plt.scatter(df['Rating'], df['Installs'], alpha=0.3)
plt.title('Rating vs Number of Installs')
plt.xlabel('Rating')
plt.ylabel('Installs')
plt.show()
```

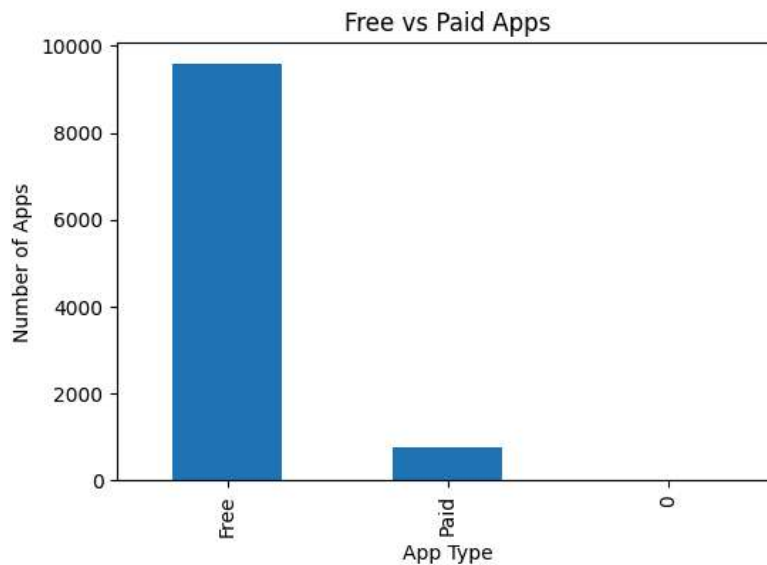


```
plt.figure(figsize=(8,5))
plt.scatter(df['Rating'], df['Installs'], alpha=0.3)
plt.yscale('log')
plt.title('Rating vs Number of Installs')
plt.xlabel('Rating')
plt.ylabel('Installs (Log Scale)')
plt.show()
```



```
df['Type'].value_counts().plot(kind='bar', figsize=(6,4))

plt.title('Free vs Paid Apps')
plt.xlabel('App Type')
plt.ylabel('Number of Apps')
plt.show()
```



```
top10 = df[['App', 'Installs']].sort_values(
    by='Installs', ascending=False
).head(10)
```

top10

	App	Installs	
3234	Google	1.000000e+09	
2554	Google+	1.000000e+09	
2545	Instagram	1.000000e+09	
3736	Google News	1.000000e+09	
5395	Google Photos	1.000000e+09	
2544	Facebook	1.000000e+09	
2884	Google Photos	1.000000e+09	
3117	Maps - Navigate & Explore	1.000000e+09	
1654	Subway Surfers	1.000000e+09	
3687	Google Play Movies & TV	1.000000e+09	

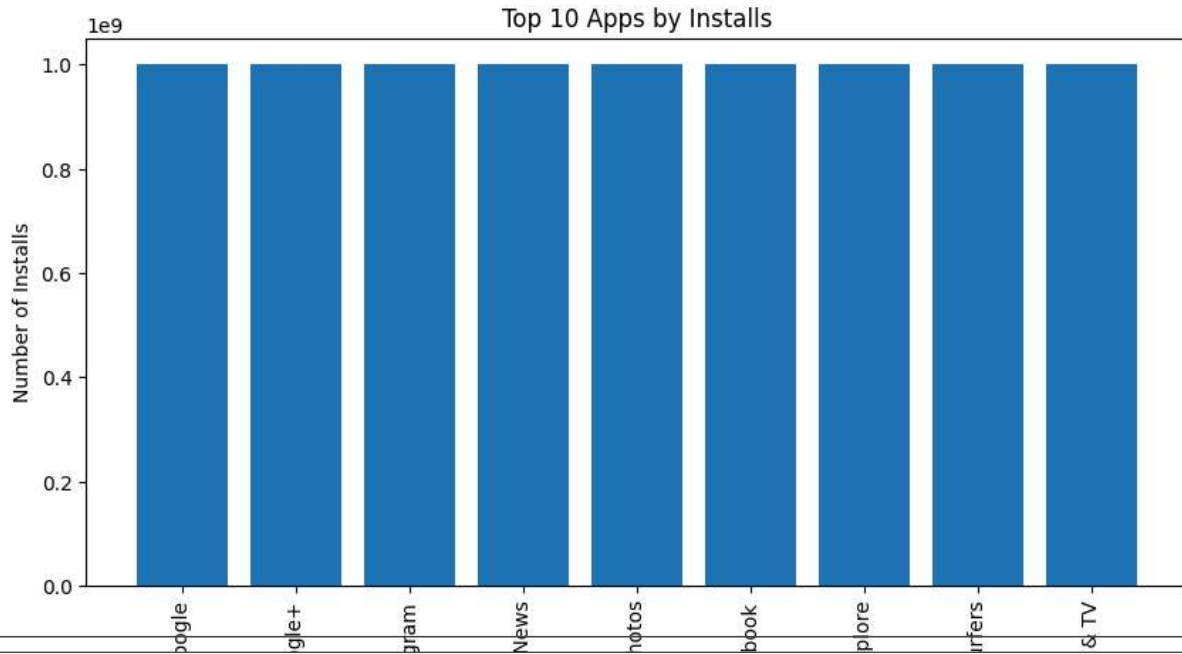
Next steps: [Generate code with top10](#) [New interactive sheet](#)

```
plt.figure(figsize=(10,5))

plt.bar(top10['App'], top10['Installs'])

plt.xticks(rotation=90)
plt.title('Top 10 Apps by Installs')
plt.xlabel('App')
plt.ylabel('Number of Installs')

plt.show()
```



```
plt.figure(figsize=(8,5))

df.groupby('Category')['Installs'].sum().sort_values(ascending=False).head(10).plot(kind='bar')

plt.title('Top Categories by Total Installs')
plt.xlabel('Category')
plt.ylabel('Total Installs')
plt.xticks(rotation=45)

plt.show()
```

